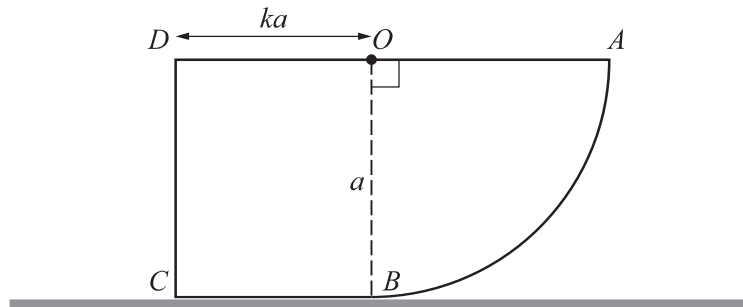


- 3 A uniform lamina $OABCD$ is in the form of a rectangle, $OBCD$, joined along the edge OB to a quarter circle OAB . The length of DO is ka and the length of OB is a . The lamina rests in a vertical plane with its edge CB on a horizontal surface (see diagram).



- (a) Find, in terms of k , a and π , an expression for the distance of the centre of mass above the horizontal surface. [4]

[You may use without proof the result for the centre of mass of a circular sector in the list of formulae (MF19).]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.



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The lamina is on the point of toppling about B .

- (b) Find the value of k . [2]

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